

# DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,  
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001 – 2015 Certified Institution)  
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560 111, India

## DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



### INTERNSHIP REPORT ON

**“Empowering Data Engineering: Automation, Analytics, and Innovation”**

*Submitted in partial fulfillment for the award of the degree of*

**Bachelor of Engineering**

**In**

**Artificial Intelligence and Machine Learning**

Submitted by:

**Shashank L S - 1DS22AI406**

Undertaken at



4th Floor, Kabra Excelsior, 80 Feet Rd, Koramangala 1A Block, Koramangala 3 Block,  
Koramangala, Bengaluru, Karnataka 560034

**Department of Artificial Intelligence and Machine Learning,  
Dayananda Sagar College of Engineering, Bengaluru**

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## CERTIFICATE

This is to certify that the Internship titled **“Empowering Data Engineering: Automation, Analytics, and Innovation”** carried out by **Shashank L S [1DS22AI406]**, a bonafide student of **Dayananda Sagar College of Engineering**, in partial fulfillment for the award of **Bachelor of Engineering in Artificial Intelligence and Machine Learning** during the academic year **2024–2025**. It is certified that all corrections/suggestions indicated have been incorporated in the report. The internship report has been approved as it satisfies the academic requirements in respect of the course Internship (21INT82).

Signature of the  
Reporting Manager  
Bharat Chandra Obili,  
Product Manager,  
Ai palette

Signature of the  
Internship Coordinator  
Prof. Kavya D N,  
Assistant Professor,  
Dept. of AI & ML, DSCE

Signature of the  
Head of the Department  
Dr. Vindhya P Malagi,  
Professor & Head,  
Dept. of AIML, DSCE

**Name of the Examiners**

**Signature with date**

1. ....

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2. ....

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## DECLARATION

I, **Shashank L S**, a final year student of **Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bengaluru**, hereby declare that this internship report is based on the work undertaken during my ongoing internship in the role of **Intern, Data Strategy and Automation** at **AI PALETTE INDIA PRIVATE LIMITED, Bengaluru**.

This report is submitted in partial fulfillment of the requirements for the award of the Bachelor's Degree in Artificial Intelligence and Machine Learning during the academic year 2024–2025.

**Date: 12/04/2025**

**Place: Bengaluru**

## ACKNOWLEDGEMENT

This internship is the result of continuous guidance, support, and encouragement from several individuals, to whom I am deeply grateful. I take this opportunity to express my sincere appreciation to everyone who contributed to the successful completion of this internship.

I would like to express my heartfelt thanks to **Dr. B G Prasad**, Principal, Dayananda Sagar College of Engineering, for providing the necessary facilities and support to undertake this internship.

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I am thankful to **Prof. Kavya D N**, Internship Coordinator and Assistant Professor – AI & ML, for her continuous support, timely suggestions, and motivation during the internship period.

I would also like to extend my thanks to **Dr. Aruna M G**, internal guide, for her consistent guidance, timely feedback, and unwavering support throughout the internship. Her mentorship helped me stay aligned with both academic and industry expectations.

I would like to thank **Bharat**, Product Manager, AI Palette, for the valuable industry insights and mentorship provided during my role as an intern in Data Strategy and Automation Intern.

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**SHASHANK L S [1DS22AI406]**

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# **CHAPTER 1**

## **INTRODUCTION**

In the competitive landscape of the retail industry, customer retention and loyalty have emerged as critical factors influencing long-term business sustainability. With rising customer acquisition costs and shifting consumer behaviour, businesses are increasingly turning to data analytics to gain insights into purchasing patterns, customer preferences, and churn risks. Effective customer retention strategies not only enhance revenue growth but also foster stronger brand relationships.

This project focuses on the analysis of customer behaviour using various marketing metrics such as acquisition cohort size, email subscriber engagement, Average Order Value (AOV), and Units per Transaction (UPT). These indicators are instrumental in understanding the effectiveness of marketing campaigns and in identifying high-value customer segments. Analysing these metrics helps in measuring customer lifetime value and in developing personalized marketing interventions.

Cohort analysis forms a core component of this project, offering a structured approach to track the behaviour of customer groups over time based on their acquisition period. This method provides deeper insights into customer engagement trends, retention rates, and the long-term value of different customer segments. By comparing cohorts, the project identifies drop-off patterns and the influence of external factors on customer retention.

To complement the retention analysis, a predictive model for customer churn has been developed. The model utilizes historical transaction data and behavioural attributes to identify customers at risk of disengagement. Key features used include purchase frequency, recency, and order value trends. By applying classification algorithms, the model aims to support proactive decision-making in customer relationship management and reduce churn rates through targeted marketing strategies.

The overall objective of the project is to leverage marketing analytics to drive data-informed decision-making in the retail sector. By combining descriptive and predictive techniques, the project contributes to optimizing retention strategies, enhancing customer loyalty, and improving overall marketing efficiency.

## CHAPTER 2

### ABOUT THE COMPANY



Ai Palette is an innovative AI-powered consumer insights platform that is transforming the landscape for Consumer Packaged Goods (CPG) brands. By leveraging a wealth of data from around the globe, the company provides actionable, real-time intelligence that drives smarter product development, marketing strategies, and overall business growth. This report offers an in-depth look at the company's journey and the impactful technologies it employs.

At the heart of Ai Palette's success is its commitment to harnessing advanced analytics and machine learning. The platform continuously monitors emerging trends and evolving consumer behaviors, turning vast amounts of data into accessible insights. This capability not only provides clarity in an increasingly complex market but also enables brands to proactively anticipate and adapt to shifts in consumer preferences.

The company's robust infrastructure is built on scalable data engineering solutions that integrate cutting-edge tools and methodologies. From efficient data pipelines and powerful search indexing to automated validation processes using large language models (LLMs), Ai Palette demonstrates how innovative data strategies can deliver a competitive edge. These practices ensure that the data remains reliable, accurate, and actionable, fueling informed decision-making across various business functions.

This report serves as an introduction to the strategic approach and technological prowess that underpin Ai Palette's operations. It will detail the company's methodologies, the challenges it has encountered along the way, and the innovative solutions that have been implemented. Through this exploration, the report aims to provide valuable insights into how Ai Palette is setting new benchmarks in data engineering and consumer insights, ultimately driving data-driven transformation within the CPG industry.

## **CHAPTER 3**

### **SCOPE OF THE WORK**

#### **1. Data Pipeline Architecture & Engineering:**

This work involves designing and developing robust, scalable data pipelines that ingest, transform, and index large volumes of data from various sources. The goal is to create a reliable data foundation that supports real-time insights and enhances strategic decision-making for the company. By implementing best practices in data engineering, the pipelines ensure data is cleaned, validated, and stored efficiently, enabling seamless flow from acquisition to actionable outcomes.

#### **2. Data Automation & Processing Optimization:**

A significant aspect of the work focuses on automating repetitive tasks and optimizing processing time to ensure efficiency across data workflows. This includes refining codebases, standardizing workflows, and incorporating containerization where needed. The objective is to reduce manual interventions, cut down on processing delays, and ensure that data is processed quickly and accurately, which ultimately contributes to lower operational costs and faster turnaround times.

#### **3. Data Visualization & Reporting:**

The scope extends to developing dynamic dashboards and real-time reporting tools using visualization platforms. This involves configuring tools like Kibana to build interactive dashboards that facilitate easy monitoring of key performance indicators and system performance. The visualizations help stakeholders access complex data insights quickly, supporting operational transparency and data-driven decision-making across various business functions.

#### **4. Integration of ML Models & LLM-based Data Validation:**

This component leverages advanced machine learning models and large language models (LLMs) to automate data validation and enhance predictive analytics. By integrating these models within the data pipelines, the work ensures that the data is not only accurate but also enriched with predictive insights. This integration helps in identifying trends, automating error detection, and maintaining high data quality, which is critical for reliable analysis and strategy development.



## **CHAPTER 4**

### **LEARNING OBJECTIVES**

#### **1. Develop Expertise in Data Pipeline Architecture**

Gain hands-on experience in designing and constructing robust, scalable data pipelines. This objective focuses on understanding the end-to-end data flow—from ingestion, transformation, to indexing—ensuring that you can build systems that manage vast datasets efficiently and reliably. Mastery in this area will enable you to support real-time analytics and facilitate seamless data processing across different platforms.

#### **2. Enhance Skills in Data Automation & Workflow Optimization**

Focus on automating complex data workflows to reduce manual intervention and improve processing speed. This learning objective aims to deepen your understanding of automation strategies, code optimization, and the use of containerization where applicable. Developing these skills is critical for streamlining operational processes and ensuring timely data delivery, which directly impacts decision-making efficiency.

#### **3. Master Data Visualization & Reporting Tools**

Learn to develop interactive dashboards and real-time reporting mechanisms using advanced visualization platforms. This objective emphasizes using tools like Kibana to create intuitive and insightful visualizations, enabling stakeholders to monitor key performance metrics effectively. Mastery in data visualization helps bridge the gap between raw data and actionable insights by translating complex analyses into easily digestible formats.

#### **4. Integrate Machine Learning & LLMs for Data Validation**

Expand your expertise by incorporating machine learning models and large language models (LLMs) to automate and enhance data validation processes. This learning objective focuses on using advanced algorithms to ensure high data quality, predict trends, and automate error detection. By integrating these cutting-edge techniques, you will enhance the accuracy and reliability of data analytics, ultimately supporting smarter business decisions.

## **CHAPTER 5**

### **CHALLENGES AND SOLUTIONS**

#### **1. Dealing with Incomplete or Noisy Data**

One of the major challenges encountered during the project was the presence of incomplete, inconsistent, or noisy customer data. Missing values in key fields such as transaction dates or order values affected the accuracy of retention and cohort analyses. To address this issue, various data cleaning techniques were applied, including imputation of missing values, outlier detection, and filtering of irrelevant records. Ensuring data quality was critical before proceeding with any form of analysis or modeling to maintain the reliability of results.

#### **2. Identifying Relevant Features for Churn Modeling**

Building a churn prediction model required careful selection of input features that were both predictive and interpretable. With numerous customer attributes and behavioral metrics available, identifying the most relevant features posed a significant challenge. This was addressed through a combination of exploratory data analysis (EDA), correlation heatmaps, and feature importance rankings from preliminary machine learning models. Domain knowledge was also applied to prioritize features that directly reflected customer engagement and purchase behavior.

#### **3. Balancing Imbalanced Classes in Churn Dataset**

The churn dataset was naturally imbalanced, with a smaller proportion of customers labeled as "churned" compared to those who remained active. This imbalance risked skewing the model's performance toward the majority class. To mitigate this, resampling techniques such as Synthetic Minority Over-sampling Technique (SMOTE) and stratified sampling were used to balance the dataset. Additionally, evaluation metrics like F1-score and ROC-AUC were preferred over simple accuracy to better assess model effectiveness in handling class imbalance.

#### **4. Translating Technical Insights into Business Language**

Another challenge was effectively communicating analytical insights and technical findings to non-technical stakeholders, such as marketing and business teams. This was crucial in ensuring that the insights could be understood and acted upon. To address this, the results were presented using visual dashboards, summary tables, and clear business-focused narratives.

## **CHAPTER 6**

### **ACHEIVEMENTS AND CONTRIBUTION**

#### **1. Optimizing Bitbucket Code Processing Time**

One of the primary challenges encountered was the lengthy processing time in Bitbucket during code commits and merges. These delays impacted the deployment cycles and slowed down the overall development process. To address this, I conducted an in-depth analysis of the existing codebase to identify bottlenecks and inefficient processes. By refactoring critical modules and streamlining the workflow with enhanced automation scripts, the processing time was significantly reduced. This improvement not only accelerated development cycles but also increased team productivity by enabling faster testing and deployment of new features.

#### **2. Production Data Validation with LLM Integration**

Maintaining the accuracy and reliability of production data was a critical challenge that initially relied heavily on manual checks. This approach was not only time-consuming but also prone to human errors and inconsistencies. By integrating large language models (LLMs) into the validation process, I was able to automate the error-detection mechanisms. The LLM-based system provides an additional layer of scrutiny by cross-checking data entries and flagging anomalies in real time. This solution has enhanced data quality, reduced the incidence of errors, and provided a scalable method for validating large volumes of data without compromising on accuracy.

#### **3. Automating Complex Data Workflows**

The complexity and variety of data workflows posed significant challenges in terms of efficiency and consistency. Manual intervention in these processes often led to delays, increased the risk of errors, and made it difficult to ensure a uniform approach to data handling across multiple pipelines. To overcome this, I implemented end-to-end automation solutions that standardized and optimized each step of the workflow. This included automating data ingestion, transformation, and indexing, which reduced manual workload and minimized processing times. The automated workflows have streamlined operations, allowing for real-time updates and faster turnaround from raw data to actionable insights, ultimately supporting more agile decision-making.

#### **4. Standardizing Data Translation Across Multiple Languages**

Managing and integrating data from diverse geographical regions, each with its own language, was another significant challenge. Inconsistent translations and varying data formats created barriers to achieving a unified view of the data, complicating cross-border analysis and reporting. By deploying advanced machine translation techniques and integrating language-specific models, I was able to standardize the translation process. This solution ensured that data from multiple languages was normalized and integrated seamlessly into a centralized system. The consistent translation approach not only improved the accuracy of the data but also facilitated better global insights, enabling the organization to make well-informed decisions across different markets.

## **5. Integrating Heterogeneous Data Sources**

One of the ongoing challenges was integrating data from diverse sources, each with its own structure, format, and interface. These differences made it difficult to create a uniform and seamless data flow within the pipelines. To overcome this, I developed customized adapters and standardized data mapping techniques that helped harmonize various data formats. This solution not only improved data consistency but also ensured that the integrated data was reliable and ready for further processing and analysis.

## **6. Ensuring Robust Security and Compliance**

Maintaining data security and ensuring compliance with industry regulations was another significant challenge, particularly in handling sensitive consumer insights. The complexity increased when data was processed across multiple platforms and regions with varying security standards. I addressed this by implementing comprehensive security protocols, including encryption, access control, and regular audits. These measures ensured that data remained protected at every stage of the processing pipeline while upholding regulatory requirements, thus fostering trust and integrity in the overall data management system.

## **CHAPTER 7**

### **REFLECTION AND ANALYSIS**

The internship experience has been both enriching and transformative, allowing me to bridge the gap between academic knowledge and practical application. Working on real-world customer data gave me the opportunity to apply statistical concepts and machine learning techniques in a meaningful way. I was able to observe how data-driven decisions directly influence marketing strategies and business growth, particularly in a fast-paced retail environment.

One of the most valuable lessons I learned was the importance of data quality and interpretation. While working on churn modeling and retention analysis, I realized that data cleaning and preprocessing were as crucial as model development. Making sense of incomplete, noisy, or inconsistent data challenged my problem-solving skills and taught me to be meticulous and detail-oriented. It also highlighted the significance of choosing the right metrics and features for any analytical task.

Throughout the internship, I became more confident in using tools such as Excel, Python, and data visualization libraries. Creating dashboards and reports taught me the importance of communicating insights in a clear and actionable format. I learned that even the most accurate model loses its value if the results are not understood by decision-makers. This experience helped me grow not only technically but also in terms of communication and business thinking.

Overall, the internship reinforced my interest in the field of marketing analytics and made me aware of the critical role data plays in shaping customer experiences and business strategies. It gave me a sense of responsibility, encouraged collaboration with cross-functional teams, and motivated me to continuously learn and improve. This journey has laid a strong foundation for my future career in analytics and has given me the confidence to tackle real-world challenges.

## **CHAPTER 8**

### **RECOMMENDATIONS**

Reflecting on my journey in data engineering, I have witnessed significant professional growth, acquiring a deeper understanding of designing and optimizing scalable data pipelines. This experience has not only honed my technical skills but also instilled in me a greater appreciation for the importance of system architecture in supporting real-time, data-driven decision-making. The iterative process of development and continuous improvement has been a rewarding challenge that has expanded my perspective on tackling complex data systems.

The integration of machine learning and large language models (LLMs) into our data validation processes stands out as a pivotal achievement. This initiative has transformed traditional manual checks into an automated, intelligent system that ensures high data quality and reliability. Analyzing the improvements brought by these technologies has reinforced the value of adopting cutting-edge tools in our workflows, ultimately enhancing productivity and reducing error rates across our data pipelines.

Navigating the complexities of data integration and multi-language translation has provided invaluable insights into the challenges of handling heterogeneous data sources. The adoption of advanced machine translation techniques and standardized mapping has not only resolved language inconsistencies but has also enabled a unified approach to global data analysis. This reflection on overcoming linguistic and structural disparities has highlighted the importance of adaptable solutions when dealing with diverse datasets and international data standards.

Finally, ensuring robust security and compliance has been a critical aspect of the project that demanded continuous evaluation and enhancement of our data management practices. Implementing stringent security protocols and adopting comprehensive auditing mechanisms have safeguarded our systems against vulnerabilities while meeting regulatory requirements. This analytical reflection underscores the significance of balancing operational efficiency with stringent security measures, paving the way for future improvements in data governance and system resilience.

## **CHAPTER 9**

### **CONCLUSION**

In conclusion, this project has demonstrated the transformative impact of a robust data engineering framework on strategic decision-making. By building scalable data pipelines and integrating advanced automation techniques, we have successfully transformed disparate data sources into actionable intelligence. This approach has not only streamlined the data flow but also enhanced our ability to extract insights in real time. The project validates the critical role of strong data infrastructure in responding to rapidly changing business environments and underscores the importance of continuous innovation in data management practices.

The integration of machine learning models and large language models (LLMs) for predictive analytics and data validation has markedly improved the quality and reliability of our data. These advanced technologies have shifted our data validation processes from manual, error-prone tasks to automated, intelligent systems that can detect anomalies and ensure consistency. This shift has significantly reduced processing errors and increased confidence in the insights generated. The successful implementation of these models serves as a compelling case for leveraging AI to automate complex processes and drive operational efficiency.

Addressing challenges related to heterogeneous data integration, processing delays, and multilingual data translation has been integral to the project's success. By developing standardized mapping techniques and deploying machine translation solutions, we have overcome barriers that previously hindered the consolidation of global datasets. These solutions have ensured consistent data quality and allowed us to perform comprehensive analysis across diverse sources. The lessons learned from resolving these challenges emphasize the need for flexible, adaptive solutions that cater to the intricacies of a global data landscape.

Overall, this project has cemented the importance of advanced data engineering and automation in maintaining a competitive edge in the digital era. The enhancements in data processing, validation, and integration have not only improved our operational efficiency but have also empowered strategic, data-driven decision-making across the organization. The success of this project lays a strong foundation for further innovations, ensuring that our data infrastructure remains agile, resilient, and capable of driving continuous improvement and long-term growth.